Gluing Trace-Compatible Markov Chains

Proofs for independent checking · Robert D. McRae · Oct 11, 2026

Summary

We ask when finite Markov chains that agree on their overlaps (equal traces, i.e. censored chains, on shared states) can all be realised as watched processes of one finite Markov chain, allowing several states of that chain to share a label.

- **Gluing.** Two agreeing chains always can (Theorem U1). So can any family whose overlaps form a tree (Theorem T, via Lemma L, which glues a new chain onto any existing master, hidden states included).
- **Loops.** For a loop of agreeing chains, a product of stationary-weight ratios must equal 1 (Theorem U2). That is not sufficient (Theorem U3). A quantitative version (Theorems U4, U5) bounds the "openness" of one chain in a loop by the near-determinism of the others: $O \leqslant 4e/(1-e)$.

What we ask of the reader. Please check the proofs, especially U1 step 5, Lemma L steps 2–3 and the U5 limit argument. Counterexamples are as welcome as confirmations.

Status. All proofs are by the author with AI assistance (Claude). Two adversarial checks by separate AI agents found a wrong definition and several gaps, now repaired, and no counterexample. Exact numerical checks pass to about 1e-15. No human mathematician has checked them yet.

1 Setting and definitions

Trace. Let Q be an irreducible stochastic matrix on a finite set V, $S \subseteq V$ nonempty, $U = V \setminus S$. The trace (watched, censored or induced chain) is $T_S = Q_{SS} + Q_{SU}(I - Q_{UU})^{-1} Q_{US}$. It is irreducible, its stationary law is Q's restricted to S and renormalised, and $\mathrm{Tr}_R(T_S) = T_R$ for $R \subseteq S$ (Crabtree–Haynsworth quotient formula).

Agent. An irreducible stochastic matrix M on a finite window S. Its stationary law is μ, strictly positive on S.

Agreement. Agents M_i on S_i and M_j on S_j agree if $\mathrm{Tr}_{(S_i \cap S_j)}(M_i) = \mathrm{Tr}_{(S_i \cap S_j)}(M_j)$.

Master. A master for agents M_i on windows S_i is a finite irreducible Markov chain P on Ω with a label map $\ell: \Omega \to \cup_i S_i$ such that, for every i: start P from its stationary law restricted to $\{\ell \in S_i\}$ (renormalised) and record ℓ only at times when $\ell \in S_i$; the recorded sequence has the law of the stationary Markov chain M_i, i.e. every finite word has exactly M_i's stationary probability. Equivalently, the chain induced by P on $\{\ell \in S_i\}$, run at stationarity, emits stationary M_i through ℓ. Every state carries exactly one label; several states may share a label ("hidden states"); P is first-order Markov on Ω.

Note on the start. Starting P at its full stationary law and recording from time 0 is not equivalent: the first recorded label is then biased, even with no hidden states (for the 6-cycle 1→3→5→2→4→6 watched on {1,2,5,6}, the first label is 5 or 6 with probability 1/3 each, not 1/4). Every state with a visible label has positive stationary weight, so an event of probability 1 from the start above has probability 1 from every such state.

Share holonomy. For agents on a loop $S_1, \ldots, S_k$ with nonempty overlaps $C_i = S_i \cap S_{(i+1)}$ (indices mod k), let $r_i = \mu_i(c)/\mu_{(i+1)}(c)$ for $c \in C_i$, and $h = r_1 \cdots r_k$.

Excursions. For M on $A = C \cup X$ and $c \in C$, an excursion from c is the path $c, x_1, \ldots, x_m, c'$ ($m \geqslant 0$, $x_j \in X$, $c' \in C$) up to the first return to C. **Fact E** (strong Markov): a path of M from c is a concatenation of independent excursions, each starting where the previous landed, and conversely; summing excursion probabilities landing at c' gives $\mathrm{Tr}_C(M)(c, c')$.

2 Gluing

Theorem U1 (two agents glue)

Let M_A on $A = C \cup X$ and M_B on $B = C \cup Y$ be irreducible, C nonempty, X, Y, C disjoint, and $\mathrm{Tr}_C(M_A) = \mathrm{Tr}_C(M_B) = N$. Then a master exists with at most $|C|(1 + |X| + |Y|)$ states. It is not unique.

Proof.

1. Path law. At $c \in C$ draw an A-excursion from c, landing at c'. Then draw a B-excursion from the same c, conditioned to land at the same c' (possible: $N(c, c') > 0$). Emit c, the X-part, the Y-part; continue from c' with fresh draws. Each pass through C is one shared step, seen by A with an X-part and by B with a Y-part.
2. Watched on A: deleting Y-symbols leaves independent A-excursions, i.e. M_A's path law (Fact E).

3. Watched on B: the landing point has law $N(c, \cdot)$, equal to B′s own landing law, and given c' the Y-part has B′s conditioned excursion law. Since $\Sigma_{c'} N(c, c')\, P_B(\text{path} \mid \text{lands at } c') = P_B(\text{path})$, this is B′s excursion law, hence M_B′s path law.
4. Markov realisation. Let $h_{c'}(y) = P_y^{(M_B)}(\text{first C-state is } c')$. States: c; (x, c) [A at x, excursion began at c]; (y, c′) [B at y, bound for c′]. From c or (x, c) with A at p: to (x2, c) w.p. $M_A(p, x2)$; A lands at c′ w.p. $M_A(p, c')$, then at once to c′ w.p. $M_B(c, c')/N(c, c')$ or to (y, c′) w.p. $M_B(c, y)\, h_{c'}(y)/N(c, c')$. From (y, c′): to (y2, c′) w.p. $M_B(y, y2)\, h_{c'}(y2)/h_{c'}(y)$; to c′ w.p. $M_B(y, c')/h_{c'}(y)$. Rows sum to 1 because $M_B(c, c') + \Sigma_y M_B(c, y)\, h_{c'}(y) = N(c, c')$ and h is harmonic.
5. Keep the states reachable from C (dropping e.g. (y, c′) with $h_{c'}(y) = 0$); they form one irreducible class. (a) Every kept state is reached from some $c \in C$, by definition. (b) Every kept state reaches C: from (x, c) the A-part follows M_A, which is finite and irreducible and so hits C with probability 1; at that step the chain enters c′ or some (y, c′) with $h_{c'}(y) > 0$, and from (y, c′) the h-transformed walk reaches c′ with probability 1, since its probability of any Y-path ending at c′ sums to $h_{c'}(y)/h_{c'}(y) = 1$. (c) The C-states communicate: summing over the intermediate states, the master moves from c to its next C-state c′ with total probability $N(c, c')$, and N is irreducible. So every kept state reaches every C-state and, through it, every kept state. From a C-start the watched processes are M_A, M_B by 2–3. Stationary start, word by word. Let G be the chain induced by the master on its B-labelled states; it is finite and irreducible, with stationary law ρ equal to the master′s stationary law restricted and renormalised. For a word $w = (b_0, \ldots, b_L)$ in B, let $f_w(\nu)$ be the probability that G started from law ν emits w as its first L + 1 labels; f_w is linear in ν. From δ_c, G′s label sequence is the Markov chain M_B from c (steps 3–4), so its label at step n has law $\delta_c M_B^n$, and by the Markov property of G, $f_w(\delta_c G^n) = (\delta_c M_B^n)(b_0)\, \Pi_t M_B(b_{t-1}, b_t)$. Average over $n = 0, \ldots, m - 1$: by linearity, $f_w(\nu_m) = (\lambda_m)(b_0)\, \Pi_t M_B(b_{t-1}, b_t)$, with $\nu_m = (1/m)\, \Sigma\, \delta_c G^n$ and $\lambda_m = (1/m)\, \Sigma\, \delta_c M_B^n$. For finite irreducible chains (periodic or not) these Cesàro averages converge to the stationary laws: $\nu_m \to \rho$ and $\lambda_m \to \mu_B$. By continuity of f_w, $f_w(\rho) = \mu_B(b_0)\, \Pi_t M_B(b_{t-1}, b_t)$, the stationary M_B probability of w. The same argument gives M_A. QED

Example: M_A = 1→3→2→4→1, M_B = 1→5→2→6→1, C = {1, 2}. The construction gives 1→3→5→2→4→6→1; 1→5→3→2→6→4→1 is a different master for the same pair. Picking an A- or B-excursion at random at each visit to C fails (A would see 1→2 directly).

Lemma L (glue a leaf onto any master)

Let P on Ω (finite, irreducible, labels ℓ) be a master for a family F whose windows lie in a label set U. Let $C \subseteq U$ be nonempty and suppose P′s C-watched process (in the sense of the definition) is stationary Markov N. Let M_B be irreducible on $B = C \cup Y$ with $Y \cap U = \varnothing$ and $\mathrm{Tr}_C(M_B) = N$. Then $F \cup \{M_B\}$ has a master with at most $|\Omega_C|(1 + |Y|) + |\Omega \setminus \Omega_C|\cdot|C|$ states (Ω_C = C-labelled states).

Construction. States: $w \in \Omega_C$; (w, c) for $w \notin \Omega_C$, c = label of the last C-labelled state; (y, w') for B at y bound for the C-labelled state w′. From w or (w, c) ($c = \ell(w)$ if $w \in \Omega_C$): P moves to w2 w.p. $P(w, w2)$; if $w2 \notin \Omega_C$ go to $(w2, c)$; if $w2 \in \Omega_C$ with label c′, go to w2 w.p. $M_B(c, c')/N(c, c')$ or to $(y, w2)$ w.p. $M_B(c, y)\, h_{c'}(y)/N(c, c')$. From (y, w'), $c' = \ell(w')$: to $(y2, w')$ w.p. $M_B(y, y2)\, h_{c'}(y2)/h_{c'}(y)$, to w′ w.p. $M_B(y, c')/h_{c'}(y)$. Keep states reachable from Ω_C. Every reachable move has $N(c, c') > 0$, since it has positive probability in P′s C-watched process.

Proof.

1. Strong lumping: the map $(w, c) \mapsto w$, with Y-states deleted, projects the new chain onto P from any start (the P-move from (w, c) does not depend on c, and each Y-detour ends at its target). So every window of F sees exactly what it saw in P.
2. Start from $\nu = \pi_P$ restricted to Ω_C and renormalised. The C-labels are then P′s stationary C-watched process: Markov N from its stationary law ν_N (which is μ_B restricted to C and renormalised). The Y-blocks: entering w2 (label c′) from an excursion begun at c, the probability that the inserted block is $y_1, \ldots, y_m$ is, by telescoping the h-factors, $[M_B(c, y_1)\, h_{c'}(y_1)/N(c, c')] \cdot [M_B(y_1, y_2)\, h_{c'}(y_2)/h_{c'}(y_1)] \cdots [M_B(y_m, c')/h_{c'}(y_m)] = M_B(c, y_1)\, M_B(y_1, y_2) \cdots M_B(y_m, c')/N(c, c') =: q(y_1 \ldots y_m \mid c, c')$, B′s excursion law conditioned on landing at c′. Given the whole P-path, the blocks are drawn with fresh randomness and depend only on their two C-endpoints. Identity: for every word $w = c_0\, Y^1\, c_1 \cdots Y^k\, c_k$ that ends at a C-label (each Y^j a finite string in Y, possibly empty), $P_{new}(w) = \nu_N(c_0) \prod_j N(c_{(j-1)}, c_j)\, q(Y^j \mid c_{(j-1)}, c_j) = \nu_N(c_0) \prod_j M_B(c_{(j-1)}, Y^j, c_j)$, where the last factor is the M_B-probability of that excursion path; this is M_B′s probability of w from ν_N (Fact E). A word ending inside a Y-block is the disjoint union of its extensions to the next C-label, which occurs with probability 1 under both laws, so the identity extends to all finite words. (From a single hidden C-state this can fail: two copies of one label may have different next C-labels.)
3. Stationary start on B: as in U1 step 5, with the induced chain G on B-labelled states started from ν. By step 2 and the Markov property, $f_w(\nu\, G^n) = (\nu_N\, M_B^n)(b_0) \prod_t M_B(b_{(t-1)}, b_t)$; Cesàro averages give $\nu_m \to \rho$ and ν_N-averages $\to \mu_B$, so

f_w(ρ) is the stationary M_B probability of w.

4. Irreducibility: kept states are reached from Ω_C, Y-detours end in Ω_C, P is irreducible.
5. Size: |Ω_C| + |Ω \ Ω_C||C| + |Y||Ω_C|; for P = M_A this is the U1 bound. QED

Theorem T (trees glue)

Let M_1, …, M_n on S_1, …, S_n be ordered so that for k ⩾ 2, C_k = S_k ∩ (S_1 ∪ ⋯ ∪ S_(k−1)) is nonempty and contained in one earlier window S_j. If Tr_(C_k)(M_k) = Tr_(C_k)(M_j) for all k, a master exists with at most Π|S_i| states.

Proof. Induction with Lemma L. For a master P of M_1..M_(k−1): π_P restricted to C_k-labelled states equals π_P restricted to S_j-labelled states conditioned on the current label lying in C_k (an event fixed by the label, as C_k ⊆ S_j). So P's C_k-watched process is stationary M_j watched on C_k, i.e. stationary Tr_(C_k)(M_j) = N. Lemma L applies with Y = S_k \ C_k; each step multiplies the size by at most |S_k|. QED

Remarks. Agreement is assumed only along the tree; other pairs agree automatically (their overlap lies in some C_k ⊆ S_j). Such an order exists iff the windows form a connected acyclic hypergraph (running intersection). The condition is sufficient, not necessary.

3 Loop obstructions

All results here use only the recorded label sequences, so they hold for every master, hidden states included.

Theorem U2 (share holonomy)

Let agents on a loop S_1, …, S_k agree pairwise on nonempty overlaps C_i. Then r_i = μ_i(c)/μ_(i+1)(c) does not depend on c ∈ C_i, and if a master exists, h = r_1 ⋯ r_k = 1.

Proof. Agreement on C_i means a common trace, whose stationary law is μ_i and μ_(i+1) restricted to C_i and renormalised; so these restrictions are proportional and r_i is independent of c. For a master with stationary law π, let π(s) be the total weight of label s. The induced chain on {ℓ ∈ S_i} has stationary law π restricted, and by the definition its label marginal is μ_i, so π(s) = π(S_i) μ_i(s) for s ∈ S_i. On C_i this gives π(S_(i+1))/π(S_i) = r_i; the product round the loop is 1. QED

Counterexample with h = 8. Labels 1–6; windows {1,2,3,4}, {3,4,5,6}, {5,6,1,2}. Each agent, with states ordered (previous overlap, next overlap), is [[0, 1/2, 1/2, 0], [1/2, 1/2, 0, 0], [1/4, 0, 1/4, 1/2], [0, 0, 1/2, 1/2]]. Every overlap trace is [[1/2, 1/2], [1/2, 1/2]], so all pairs agree; μ =

(1/6, 1/6, 1/3, 1/3), r_i = 2, h = 8. Checked in exact arithmetic.

Theorem U3 (holonomy 1 is not enough)

On labels 1, 2, 3 let W1 on {1,2} and W2 on {2,3} alternate strictly and W3 on {1,3} be a fair coin. Then h = 1, all pairs agree (overlaps are single states), and no master exists.

Proof. Start the master from its stationary law restricted to label 2 (sufficient by the note in section 1). Label 2 recurs. Watched on {1,2}, exactly one 1 lies between consecutive 2s; on {2,3}, exactly one 3. So the {1,3}-recorded sequence is a run of whole blocks 13 or 31, and D_t = #1 − #3 among its first t symbols satisfies |D_t| ⩽ 1. Each 2-labelled state has positive weight in this start, so the block structure holds almost surely from every 2-labelled state, and by the strong Markov property at the first visit to label 2 it holds from any start after that visit; hence |D_t| ⩽ 2 almost surely from the {1,3} start of the definition. But from that start the recorded sequence must be stationary W3, under which D_t is a simple random walk, unbounded almost surely. Contradiction. QED

Theorems U4 and U5 (openness bound)

Let W1 = W2 = [[e, 1 − e], [1 − e, e]] on {1,2} and {2,3} (0 < e < 1) and W3 = [[1 − a, a], [a, 1 − a]] on {1,3}, with openness O = (1 − a)/a. **U5:** if a master exists, O ⩽ 4e/(1 − e). **U4:** with W3 a fair coin (O = 1), no master exists for e < 1/5; at e = 1/2 one does (i.i.d. uniform on {1,2,3}).

Proof of U5.

1. Start the master from π restricted to label 2. Watched on {1,2} it is W1 from 2; on {2,3}, W2 from 2.
2. Let N1_k, N3_k count the 1s and 3s between the k-th and (k+1)-th visits to 2 (first visit at time 0). W1 renews at each 2, so the N1_k are i.i.d. with P(N1 = 0) = e, P(N1 = j) = (1 − e)² e^(j−1) (j ⩾ 1), mean 1, variance s² = 2e/(1 − e). Likewise N3_k. The joint law of (N1_k, N3_k) is arbitrary.
3. With D_t as in U3 and T_m = Σ_(k⩽m)(N1_k + N3_k): D_(T_m) = Σ(N1_k − 1) − Σ(N3_k − 1). By Minkowski, ‖D_(T_m)‖*2 ⩽ 2V(m) s, so E[D*(T_m)²]/m ⩽ 4s².
4. Let H be the chain induced by the master on {ℓ ∈ {1,3}}: finite and irreducible. With f = +1 on 1-labelled and −1 on 3-labelled states, D_t = Σ_(s<t) f(H_s) is an additive functional of H. For a finite irreducible chain, periodic or not, the functional central limit theorem holds from every initial law, with the same mean and variance (via the Poisson equation and a martingale decomposition; e.g. Meyn and Tweedie 2009, ch. 17): t ↦ D_(⌊nt⌋)/√n converges to σ times Brownian motion. Mean and variance are read off at

the start of the definition, where the recorded sequence is stationary W3: mean 0 and $\sigma^2 = 1 + 2\,\Sigma_{(j\geqslant 1)}\,(1-2a)^j = (1-a)/a = O$ $(0 < a < 1)$. So from our start, too, $D_t/\sqrt{t} \Rightarrow N(0, O)$, and the functional limit holds.

5. By the law of large numbers for the i.i.d. $N1_k$ and the i.i.d. $N3_k$ (mean 1 each), $T_m/m \to 2$ almost surely. Anscombe's theorem (Anscombe 1952; see Gut 2009, ch. 1): if $D_t/\sqrt{t} \Rightarrow N(0, O)$, D satisfies Anscombe's condition (uniform continuity in probability: for every $\varepsilon > 0$ there is $\delta > 0$ with $P(\max_{(|t'-t| \leqslant \delta t)} |D_{t'} - D_t| > \varepsilon\sqrt{t}) < \varepsilon$ for all large t), and integer random indices satisfy $T_m/m \to \theta > 0$ in probability, then $D_{T_m}/\sqrt{T_m} \Rightarrow N(0, O)$. The indices may depend on the process in any way. Anscombe's condition follows from tightness in the functional limit of step 4. So $D_{T_m}/\sqrt{T_m} \Rightarrow N(0, O)$, and by Slutsky with $T_m/m \to 2$, $D_{T_m}/\sqrt{m} \Rightarrow N(0, 2O)$.
6. The variance of a weak limit is at most the lim inf of second moments, so $2O \leqslant 4s^2$, i.e. $O \leqslant 2s^2 = 4e/(1-e)$. QED

U4 follows: $O = 1$ needs $e \geqslant 1/5$. A first version of this proof (bound $16e/(1-e)$, threshold $1/17$) mixed two starting laws; the version above replaces it.

Reach (numerical only). Three-state masters without hidden states reach $a = (1-e)^2$, i.e. $O \approx 2e$ for small e. The true limit lies between about $2e$ and $4e/(1-e)$.

4 Context, checks and questions

Prior work. The trace and its nesting are classical (Lévy 1951; Meyer 1989; Aldous and Fill §2.7.1; Crabtree and Haynsworth 1969). Hoffman, Prakash and Chattopadhyay (2024, Appendix A.4) order kernels by "is a trace of" and ask for the join (least upper bound) of two kernels supported on the union of their states, with no hidden states; their Remark 4.10 says: "It is an open question whether solutions always exist and, if so, are unique." U1 answers a weaker version (any common master, hidden states allowed), not their question. Acyclic families glue in the marginal problem (Vorob'ev 1962; Atserias and Kolaitis 2020), and loop obstructions resemble cycle consistency (Singer 2011) and contextuality cohomology (Abramsky and Brandenburger 2011; Abramsky, Mansfield and Soares Barbosa 2012). A moderate search found no prior statement of U1–U5 or Theorem T for Markov traces. Pointers to prior statements are very welcome.

Computer checks (exact forward-algorithm word probabilities from the start in the definition; Python scripts available on request).

- U1: 14 adversarial pairs (periodic, zeros, $|C| = 1$, sparse); all words to length 6 match to 1.6e-15; the state bound is attained.

- Theorem T: 7 trees plus an adversarial set (hidden-copy overlaps, 5-window paths up to 81 states, stars, periodic, Y empty); worst error 7e-15. Control: after gluing two windows of the h = 8 loop, the third window's word probabilities differ from its agent by 0.23.
- U2: the h = 8 example in exact rational arithmetic.
- U5: a hidden-state search coupling the block counts found no master beyond the bound.

Questions for the reader.

1. Is the definition of master (start restricted to the window) the natural one, and is anything lost relative to watching from time 0?
2. Are U1 step 5 and Lemma L steps 2–3 complete as written?
3. Is the use of the start-independent CLT and Anscombe's theorem in U5 (steps 4–5) correct, and can the bound 4e/(1 − e) be improved towards 2e?
4. Are any of these results already known?

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